SLIDE-BY-SLIDE STUDY GUIDE — WEEK 5: LOCKING THE LINK

GROUP 9 · OFF-GRID VEHICLE TRACKER · REV F · 14 SHEETS · every acronym, number, and formula explained in plain language.
Each sheet: what it claims, then a decoder table of its terms, then the one sentence to say if someone asks "so what?"

Sheet 01 — Cover: "Locking the Link"

opener, ~45 s

The claim: on Sept 25 our two radios completed a round-trip verification on our own private channel — the radio backbone of the tracker now exists. The rest of the deck explains how that is possible, proves it with math and live hardware, and shows the display protection plan.

Term on slide	What it means
`round-trip traceroute`	A test command: radio A sends a request that radio B must decode and answer, and A must decode the answer. One command therefore proves BOTH directions of the radio link at once.
`Group9 channel`	Our team's private, encrypted radio channel. The name "Group9" also secretly selects the exact frequency (sheet 05 explains how).
`218 of 768 hours`	Team labor logged so far vs. the semester budget (4 people × 12 h/week × 16 weeks = 768).
`Heltec V4`	The radio device we use: a small board containing a LoRa radio chip, a GPS receiver, and a tiny screen, running the open-source Meshtastic firmware unmodified.

> *"Two radios, our own encrypted channel, verified in both directions — today we show you exactly how and why it works."*

Sheet 02 — What Makes Two Radios Able to Talk

DEO · the "contract" idea

The claim: LoRa radios have no pairing, no session, no handshake like WiFi or Bluetooth. Two radios can exchange data exactly when they independently agree on three things — frequency, modulation, encryption key. *Agreement IS the connection.* Each row of the table was verified on our hardware.

Term on slide	What it means
`SF / RG`	The two radios' callsigns (short names). SF rides in vehicle 1, RG in vehicle 2. Every screen in the system uses these names.
`GPS, 10 s`	Each radio has its own satellite receiver and computes a fresh position every 10 seconds.
`Meshtastic`	Open-source radio firmware. It handles packets, encryption, and the mesh protocol so we did not have to write a radio stack from scratch.
`ESP32(-S3)`	The microcontroller (small computer chip) inside both the radios and our display.
`SX1262`	The actual LoRa radio transceiver chip made by Semtech — the thing that physically transmits and receives.
`+30 dBm`	Transmit power. dBm = decibels relative to 1 milliwatt; +30 dBm = 1000 mW = one full watt, the US legal max for this band.
`912.875 MHz / slot 43`	The exact frequency both radios tune to. The band is divided into numbered "slots" (channels); ours is slot 43.
`LoRa`	"Long Range" — a radio modulation that encodes data as slow frequency sweeps ("chirps"), trading speed for extreme sensitivity and range.
`SF11 / 250 kHz / CR 4:5`	The modulation recipe. SF = spreading factor: each bit is spread over 2^{11} = 2048 "chips," making it slow but decodable in noise. 250 kHz = the channel width. CR (coding rate) 4:5 = for every 4 data bits, 1 extra error-correction bit is sent.
`LONG_FAST`	Meshtastic's name for this preset (SF11/250 kHz/4:5) — its default long-range setting.
`AES-256`	The standard encryption algorithm (Advanced Encryption Standard) with a 256-bit key. Both radios hold the same secret key; anyone without it hears noise.
`8.19 ms symbol / ~0.5 s per packet`	How slow SF11 is: each symbol takes 8.19 milliseconds on air, so one position update occupies the channel about half a second.
`OLED`	The small built-in screen on each Heltec radio (organic LED). Used as our "truth-teller" for the real operating frequency.

> *"No wires, no towers, no pairing — if both radios independently arrive at the same frequency, modulation, and key, they can talk. One mismatch, total silence. We proved every row."*

Sheet 03 — How the Link Happens: Locking On

DEO · the acquisition ladder + the proof

The claim: the "lock" is not a connection — it is bookkeeping. Each radio announces itself at boot; the other files it, tracks it, and once keys are exchanged can address it privately. No human involvement. The code box is the real milestone output.

Term on slide	What it means
`NodeInfo`	The self-introduction packet every radio broadcasts at boot and periodically: its name, hardware type, and its public key.
`public key`	Half of a cryptographic key pair. A radio publishes its public key so others can encrypt private messages that only it (holding the private half) can read.
`last-heard + signal quality`	What the receiver records about every node: WHEN it last heard it and HOW STRONG it was. The "1 online" counter is computed purely from the last-heard timestamp.
`direct message (DM)`	A message addressed to one specific node (vs. a broadcast to everyone). In modern Meshtastic, DMs are encrypted per-node using the exchanged public keys — impossible until step 1 has happened.
`!49b5ead4 / !49b5bb5c`	The radios' unique node IDs (derived from their hardware addresses). ead4 = SF, bb5c = RG.
`5.75 dB / 6.25 dB`	The measured signal-to-noise ratio of the traceroute request and its reply — each direction's link health in one number (sheet 06 explains the scale).
`channelIndex:0`	The message went on the primary channel (Group9), protected by the shared channel key — which is why traceroute works even before public keys exchange.

> *"Announce, file, track, address — power two configured radios and the lock forms by itself. The traceroute output is our acceptance criterion in a single line: both directions, with signal numbers."*

Sheet 04 — Staying Linked: The Heartbeat

DEO · what "online" really is + the traffic schedule

The claim: "online" is not a live connection indicator — it is arithmetic on stored timestamps. The formula: a node counts as online exactly when (now – time it was last heard) < 7200 seconds. We designed the traffic schedule so that arithmetic stays honest, after being burned by it on the bench.

Term on slide	What it means
`online(n) ⇔ t_now - t_last_heard(n) < 7200 s`	Read: "node n is online if and only if the current time minus the last time we heard it is under 7200 seconds (2 hours)." The ⇔ symbol means "if and only if" — the two sides are the same statement. That inequality is ALL that "1 online" means.
`epoch`	How computers store time: seconds counted since Jan 1, 1970. Both t_now and t_last_heard are epochs.
`10 s / 20 m (smart broadcast)`	While driving, a radio sends its position when 10 seconds AND 20 meters of movement have both passed — the product's live-tracking rate.
`60 s (parked with fix)`	A stationary radio with GPS lock beacons once a minute — presence without wasting airtime.
`telemetry heartbeat, 5 min`	A tiny status packet (battery, uptime) every radio sends every 300 s no matter what. We shortened it from the 30-minute default so idle radios keep re-stamping each other and the counters can't drift into lies.
`"72 d ago" photo`	Our bench bug: a STORED July timestamp kept resurfacing, so the screen said "signal GOOD" and "heard 72 days ago" simultaneously — both true, because signal is measured per packet while "online" is arithmetic on the stored stamp. Fixed by purging stale records and adding the heartbeat.

> *"There is no link light in a broadcast mesh — only timestamps. So we engineered the timestamps: 10 seconds driving, 60 parked, a 5-minute heartbeat always. Tested, not assumed."*

Sheet 05 — Math I: The Name Picks the Frequency

VICTOR · the equation that caused and explained our hardest bug

The claim: Meshtastic never stores a frequency. Each radio computes it from a hash of the channel NAME. Rename the channel and the radio literally retunes.

Symbol / term	What it means
`f_slot = f_start + BW/2 + [djb2(name) mod N_slots]·BW`	Read left to right: start at the bottom of the band (f_start = 902.0 MHz), move up half a channel width so the channel is centered (BW/2 = 0.125 MHz), then move up "slot number" whole channels (each BW = 0.25 MHz wide).
`djb2(name)`	A tiny text-to-number recipe (a "hash"): start with 5381, then for each character multiply by 33 and add the character. Five lines of code. Named after Daniel J. Bernstein, the famous cryptographer who posted it in the 1990s — he has nothing to do with our project; his math is just baked into Meshtastic, the way engineers use Ohm's law without Ohm.
`mod`	"Modulo" — the remainder after division. `hash mod 104` wraps the huge hash number into the range 0–103, a valid slot index.
`N_slots = 26 MHz / 250 kHz = 104`	The US band is 26 MHz wide (902–928); at 250 kHz per channel that is 104 possible slots.
`ISM band`	Industrial-Scientific-Medical: the 902–928 MHz range where unlicensed radios like ours are legal in the US (FCC rule 47 CFR §15.247).
`"LongFast" → slot 19 = 906.875`	Calibration proof: the default public channel name hashes to the PUBLISHED default US frequency — so the formula is verified against a known constant before we trust it for Group9 → slot 43 = 912.875.

"The channel name looks like a label, but it is a tuning word. Two radios with identical keys and different names sit 6 MHz apart and never hear each other — we proved that the hard way, then fixed it. The radio's own screen shows the live frequency: our truth-teller."

Sheet 06 — Math II: What 6 dB of Signal Buys

VICTOR · margin = range

The claim: our measured ~6 dB SNR at desk range leaves 23.5 dB of margin above the point where decoding fails — and every 6 dB of margin roughly doubles the free-space distance. That is how boring bench numbers translate into kilometers outdoors.

Symbol / term	What it means
dB (decibel)	A ratio on a logarithmic scale. +3 dB ≈ double, +10 dB = 10×, +20 dB = 100×. Used because radio quantities span trillions-to-one ranges.
dBm	The same scale pinned to 1 milliwatt of power. +30 dBm = 1 W; −131 dBm = 0.00000000000008 mW — the whisper our receiver can still decode.
dBi	Antenna gain relative to a perfect point radiator ("isotropic"). Our stock whip antennas ≈ 0 dBi (no focusing).
SNR	Signal-to-Noise Ratio: how far the signal stands above the background static, in dB. Positive = louder than the noise.
decode floor −17.5 dB	The LoRa superpower: at SF11 the receiver still decodes when the signal is 17.5 dB BELOW the noise (quieter than the static!). Possible because each bit is spread over 2048 chips the receiver adds back up coherently.
23.5 dB margin	Measured ~+6 dB minus the −17.5 dB floor. Roughly 3–4 "doublings" of distance in reserve.
P_rx = P_tx + G_tx + G_rx − FSPL	The link budget: received power = transmitted power + both antenna gains − free-space path loss. FSPL = $20\log_{10}(d) + 20\log_{10}(f) - 147.55$, which is just "+6 dB of loss every time distance doubles" with unit bookkeeping (d in meters, f in Hz).
161 dB budget	+30 dBm out, decodable down to −131 dBm: the total loss the link can absorb.
20 dB fade margin	Reality tax reserved for buildings, terrain, and signal fading. With it, the honest design range is 3–5 km; free space alone would suggest ~100 km, which terrain spends.
sensitivity S	The weakest signal a receiver can decode (from the SX1262 datasheet at our settings: −131 dBm).

> *"dB is a ratio, dBm pins it to a milliwatt, SNR is height above the static. We measured 6, decoding survives to minus 17.5 — 23.5 dB of margin, and margin IS range."*

Sheet 07 — Math III: Airtime Is the Budget

VICTOR · why we changed the radios' role

The claim: every packet costs ~half a second of channel time, the firmware throttles position updates above 25% channel usage, and the mesh default of repeating every packet would have burned our budget exactly when both vehicles move. Setting role CLIENT_MUTE halved usage for free.

Symbol / term	What it means
`T_sym = 2^SF / BW`	Time per symbol: 2^{11} = 2048 chips divided by 250,000 chips/second = 8.19 ms.
`T_pkt = (N_pre + 4.25 + 8 + N_payload)·T_sym`	A packet's air time is just symbol counting: preamble symbols (N_pre, what the receiver locks onto; the 4.25 is a fixed extra fraction the LoRa standard adds), 8 header symbols, then the data symbols. Comes out 0.4–0.9 s for our packets.
`ChUtil (channel utilization)`	The % of the last minute the channel carried radio energy. Shown live on every radio's screen.
`25% throttle`	Above ~25% ChUtil, Meshtastic suppresses position broadcasts — our core packet — to protect the shared band.
`flood mesh / relaying`	By default every node REBROADCASTS packets it hears (up to "hop limit" 3) so distant nodes get covered. With exactly two radios, every relay is pure waste — there is no third radio to reach.
`CLIENT_MUTE`	A Meshtastic role: hear everything, send your own, repeat nothing. Measured effect on our bench: 9–13% → ~5–6% ChUtil. Documented revert: if a third radio ever joins, both go back to CLIENT.
`2 × (0.5 s / 10 s) = 10%`	The vehicle-rate budget: two radios each sending a half-second packet every 10 s = 10% of the channel; with relaying it doubles to ~20%, brushing the throttle.

> *"Airtime is the currency. At vehicle rates the default mesh behavior spends twice what we do — and the overdraft penalty is losing the exact packets the tracker lives on."*

Sheet 08 — Live Demo I: Send It, Receive It

`CANNON · live radios in hand`

The claim: a text message crosses radio-to-radio with no phone, no tower, no internet — demonstrated live, with the slide's bench captures as backstop.

Term on slide	What it means
`canned messages`	15 preset texts stored in both radios, selectable with the user button (no typing while driving). The chain is designed so every question has its answer preset: "How are you?" → "All good"/"Need help"; "Are you injured?" → "Yes"/"No"; "Send position" → "Position updated".
`user button`	The Heltec's single input button; short presses scroll the preset list, select sends.
`Last-Heard list re-stamps`	The third photo: receiving the text updates the sender's "last heard" entry to seconds — the same bookkeeping sheet 04 explained, now visible live.
`"Send position" nuance (if asked)`	That preset is words for the HUMAN; the coordinates travel separately and automatically via the position broadcasts (or the display's request button). The preset choreographs people; the firmware moves data.

> *"Watch the message cross: 912.875 megahertz, our key, nothing else in the loop."*

Sheet 09 — Live Demo II: Reading the Radio

CANNON · the 8 screens, what each proves

Screen	What it shows and why it matters
DISTANCE	Straight-line meters from MY position to each node's last position. Pure geometry — needs BOTH ends to have GPS fixes, which is why it shows "?" indoors. Honesty, not failure.
BEARING	The compass direction (N/NE/E…) from me to each node — same geometry, other output.
POSITION	Own GPS state: satellites in view, fix/no-fix, altitude. The photo shows 14 satellites in the printed case, in a vehicle.
LAST HEARD	Per-node "seconds since it spoke" — the raw material of the online counter.
LORA	The radio contract in hardware: region/preset, THE LIVE FREQUENCY (912.875 = our slot; this line exposed our week-one bug), and ChUtil.
SYSTEM	Firmware version (2.7.26, deliberately identical on both radios) and memory stats.
TIME	GPS-disciplined clock — satellites carry atomic time; needs a fix to be right. Also feeds the timestamps behind "online."
NODE PAGE	Per-peer detail: signal bars and heard-age for one node — RG's view of SF.

> *"Short-press cycles the pages. Every number you just learned in the math sheets has a live home on this screen."*

Sheet 10 — How the Testing Was Done

CANNON · the credibility sheet

Term on slide	What it means
`one variable per trial`	Change exactly one setting, retest, log. This is what finally isolated the frequency-slot mismatch after full config comparisons showed "identical."
`A/B regression`	Retreat to a known-good configuration (the public default channel the radios had used in July), prove the hardware there, then rebuild ours. One test split the world in half: hardware fine → our channel config guilty.
`machine probes`	Tests a computer can check: traceroute (round-trip + SNR numbers) and node-database dumps over USB — instead of squinting at screens.
`read-back + OLED verify`	Every config write verified twice: read back over serial AND cross-checked on the radio's screen — because we caught writes silently reverting (the serial port's reset line can reboot the chip mid-save).
`NAK`	Negative AcKnowledgment — a delivery-failed reply. We logged every one as evidence: a NAK that ARRIVES BY RADIO proves the radio path even while the app layer refuses. (ACK = the success twin.)
`p95 ≤ 15 s`	The drive-test bar: the 95th percentile of partner-position age while moving — all but the worst 5% of samples must be fresher than 15 s. Percentiles stop one lucky (or unlucky) sample from deciding the result.
`convoy / split / rejoin`	Drive-test phases: follow close (the product experience), separate to 0.5/1/2 km (SNR vs distance data), converge (reacquisition time — bar: ≤60 s).

> *"The link works because the testing was disciplined, not lucky — and every metric was declared before the drive, so success can't be redefined afterward."*

Sheet 11 — Protecting the Display: The Case

JOSIAH · why + what

Term on slide	What it means
RLCD	Reflective LCD — a screen type that reflects ambient light instead of using a backlight. That is why it is readable in direct sun on a dashboard, and why it produces almost no heat.
18650	A standard cylindrical lithium battery size (18 mm × 65 mm) — the display board has a holder for one; the case keeps that bay reachable.
M2.5×6	Metric machine screw: 2.5 mm thread diameter, 6 mm long. Four of them hold the case, into holes in the display's own back.
Printables 1610004 / AWSW	The open 3D-model site, the model number, and its author's handle — our required credit.
CC BY-NC 4.0	Creative Commons license: free to use and remix WITH attribution, NOT for commercial sale. A class project qualifies.
zero RF shielding / BLE	The case is plastic, which is transparent to radio waves — so the display's Bluetooth Low Energy link to its radio is unaffected (a metal box would block it).
~40 g filament ≈ $1	The entire material cost of one case; one overnight print.

> *"Our first display arrived cracked and dead — the replacement rides in armor. Existing open design, one dollar of plastic, every port still reachable, invisible to the radio link."*

Sheet 12 — Print It, Screw It, Done

JOSIAH · print settings decoded

Term on slide	What it means
`0.2 mm layers`	Layer height: each deposited slice of plastic is a fifth of a millimeter — the standard strength/speed compromise.
`organic supports`	Sacrificial tree-shaped scaffolding printed under overhangs, snapped off afterwards; "organic" (tree) supports touch less surface, so less scarring than classic grid supports.
`fuzzy skin`	The printer nozzle deliberately jitters while printing the outer walls, creating a textured surface that hides layer lines and adds grip.
`no brim`	A brim is a one-layer skirt around a part that fights warping/adhesion problems; this model doesn't need one.
`PETG vs PLA`	Two common printing plastics. PLA softens around 60 °C — a Texas dashboard exceeds that. PETG holds to ~80 °C, so the case is PETG.
`sequential ("separate part") printing`	Each small part is completed before the next starts — one mid-print failure can't ruin the whole plate.
`3MF`	The modern 3D-print file format (geometry PLUS embedded print settings, unlike bare STL). The author ships "-with-settings" plates we start from.
`the ONE RULE (6 mm max)`	The screw holes are in the display's own back — longer screws press directly on the glass. The author's own warning; we highlighted it in red.
`acceptance: BLE within 2 dB, 1 m drop`	The case passes only if: no pressure on glass, all ports/buttons usable, Bluetooth signal cased vs bare differs ≤2 dB, and it survives a 1-meter drop. Measurable, like every other claim in this deck.

> *"Seven parts, one overnight run, one rule: six-millimeter screws, no substitutes."*

Sheet 13 — The Driver Display, Live on Glass

JOSIAH · the money sheet

Term on slide	What it means
`callsign header (RG)`	The display names the partner by its real radio callsign, learned automatically from NodeInfo packets — no configuration.
`"ead4<->RG OK" tag`	The link indicator: shown while the partner has been heard within 90 s. Computed from packet freshness — the sheet-04 arithmetic made visible. (ead4 was SF's temporary default name in the photo; it now reads SF.)
`the popup band`	"RG: Are you injured?" — a canned message sent from RG's button, decoded by the display and shown for 8 seconds. Text rides the same decode path as positions.
`13 ft`	Two independent GPS receivers on one table agreeing to 13 feet — excellent for consumer GPS (typical error is 10–30 ft each).
`"RG IS N OF ead4"`	The relative direction spelled out from the two fixes — the bearing math in words.
`12:03a CDT clock`	Location-aware local time: GPS supplies UTC (atomic time), the firmware derives the timezone from position — no setting, works crossing zones.

> *"Nothing on these screens is simulated — names from NodeInfo, the link tag from packet freshness, the popup from the mesh, thirteen feet from two live GPS fixes."*

Sheet 14 — The Product, On One Table

JOSIAH · closer

Term on slide	What it means
`the table shot`	Both radios in printed enclosures with antennas, the driver display between them reading 11 ft — the complete tracking chain in one photo, with zero infrastructure ("off-grid" made literal: no phone, no tower, no internet anywhere in the loop).
`"stopped"`	The display's honest label when neither vehicle is moving: with no motion there's no heading, so the compass shows absolute direction instead of turn guidance.
`age 54s`	Seconds since the partner's data arrived — the display never pretends data is fresher than it is.
`NEXT box`	Remaining work is physical: print the display's own case, then the two-vehicle drive test — this same table, stretched across town.

> *"Everything the project promised is on this table and working. Next: the same picture at highway speed."*

Master cheat-sheet (one breath each)

Term	One-breath answer
`LoRa`	Long-range chirp radio: slow on purpose, hears below the noise.
`Meshtastic`	The open-source firmware running our radios, unmodified.
`SF11`	Each bit spread over 2048 chips — slow, but decodes 17.5 dB under the noise.
`BW / CR`	Channel width (250 kHz) / error-correction ratio (4:5).
`dB / dBm / dBi / SNR`	Ratio / power vs 1 mW / antenna gain vs ideal / signal height above noise.
`AES-256 / PSK`	The encryption algorithm / the Pre-Shared Key both radios hold.
`PKC / public key`	Per-node key pairs used only for private direct messages.
`NodeInfo / DM / ACK / NAK`	Self-introduction packet / addressed message / delivered / delivery-failed.
`djb2 / mod`	Bernstein's 5-line text-to-number hash / the remainder operation that wraps it to 0–103.
`ISM / FCC / CFR`	The license-free 902–928 MHz band / US radio regulator / the rulebook (§15.247).
`ChUtil / ToA`	% of the last minute the channel was busy / one packet's time on air.
`CLIENT_MUTE`	Radio role: hear and send everything, relay nothing — right for a 2-node mesh.
`GNSS / GPS / epoch`	Satellite navigation (generic / the US system) / seconds since Jan 1 1970.
`BLE / GATT`	Bluetooth Low Energy, the display↔radio link / its data-exchange structure.
`ESP32-S3 / SX1262 / MCU / PHY`	Our processor chip / the LoRa transceiver chip / microcontroller / the physical radio layer.
`OLED / RLCD`	The radio's small screen / the display's sunlight-readable reflective panel.
`SMA / 18650 / M2.5×6 / 3MF / PETG / PLA`	Antenna connector / battery size / screw spec / print file format / heat-tolerant plastic / standard plastic.
`p95`	95th percentile — all but the worst 5% of samples must beat the bar.
`traceroute`	The round-trip test command: request out, reply back, SNR per hop — both directions proven at once.

Q&A Appendix — questions we practiced, answers ready to give

Q: "104 slots × 250 kHz" — what does that mean?

It is the parking-lot layout of the band. The lot: the US license-free band runs 902–928 MHz — a strip 26 MHz wide. The parking spaces: our LoRa signal occupies 250 kHz of width whenever it transmits (the BW in the modulation recipe), so the band is sliced into spaces exactly that wide. The count: 26 MHz ÷ 250 kHz = **104 spaces**, numbered 0–103, called frequency slots. Each slot's center frequency is arithmetic: band start + half a slot (to sit centered, not on the line) + slot × 0.25 MHz. Slot 19 → 906.875 (the public default); slot 43 → 912.875 (Group9). The two bold ticks on the sheet-05 diagram are those two spaces — 24 spaces (6 MHz) apart, which is why a radio parked in the wrong space hears nothing. Bonus: the 104 depends on the bandwidth — at 125 kHz channels the same band would hold 208 slots — which is why the formula writes N_slots = 26 MHz / BW instead of hard-coding 104.

Q: Is slot 43 a private channel?

No — the frequency is public airspace; the KEY is what is private. Anyone can tune 912.875 MHz and receive our transmissions; slot 43 buys us *quiet* (the public Lubbock mesh lives on slot 19, so our slot has essentially zero competing traffic), not secrecy. Privacy comes from the AES-256 pre-shared key: without it, a listener receives our packets as valid radio symbols but unreadable ciphertext — the "heard, not read" cell of the failure matrix. Honest caveat if pressed: packet HEADERS (sender ID, timing) are not encrypted, so an eavesdropper learns that a node transmitted and when — but not where it is or what it said. Podium line: *"The frequency is public — privacy comes from the encryption. Slot 43 gives us a quiet channel; the AES-256 key makes it a private one."*

Q: Why is the "online" window 7200 s — why not 1 hour or 1 minute?

Two-layer answer. (1) We didn't pick it — it is a constant in stock Meshtastic firmware: `#define NUM_ONLINE_SECS (60 * 60 * 2) // 2 hrs to consider someone offline`, in `src/mesh/NodeDB.cpp` of the official meshtastic/firmware repository. (2) Why they chose something that long: "online" is inferred purely from silence vs. traffic, so the window must exceed the longest LEGITIMATE silence. The broad Meshtastic ecosystem includes solar relays that sleep and battery nodes announcing every 30 minutes to 3 hours — a 1-minute window would flap them all offline constantly. Two hours means few false alarms at the cost of slow detection (a dead radio looks alive up to 2 h). Our system doesn't live with that looseness: our traffic contract guarantees a packet at least every 5 minutes, so the DRIVER DISPLAY layers stricter thresholds on top — link tag at 90 s, projection warnings (inverted age, hollow dart) at 30 s, STALE banner at 120 s. The radio's counter is the loose outer bound; our screens are the strict truth.

Q: What did the range test actually measure? (2026-10-05, N County Road 2000)

One continuous open-road session, filmed live, SF kit parked 4.1 road miles out (Find My ground truth), RG kit driving. The numbers, in order: full two-way tracking held to **3.26 miles (5.25 km)** — the status page read LINK CONNECTED with BOTH coordinate sets live, and the displayed distance matches the coordinate delta exactly (0.04719° of latitude = 3.26 mi; the haversine is checkable from the video frame). The stale edge arrived by **3.75 miles (6.03 km)**: DATA STALE banner on the display, age 2 min inverted, and "Message failed" on the radio itself — the failure announced on both devices. On the drive back the system reacquired by itself — ages back in single digits, dart at 12 o'clock at 37 mph, zero reconnects over the whole session. Design range was 3–5 km with terrain; we measured the top of it plus about a kilometer, with the antennas still INSIDE the cars. Podium line: *"We designed for 3 to 5 kilometers; the road gave us 5.25 healthy and the edge past 6."*

Q: So is 3.75 miles the limit of the communication range?

It is the measured reliable edge *for that configuration* — not a physics wall. Three qualifiers to give. (1) Configuration: antennas were inside the vehicles; a car body costs 10–20 dB, and by the 6-dB-doubles-distance rule, a cheap magnetic roof antenna alone could roughly double to quadruple the range. (2) Environment: the SAME hardware went stale at 0.5 miles in town the night before — a factor of seven from surroundings alone, which is exactly the 20 dB environmental reserve in our link budget made visible. (3) The edge is a zone, not a line: at 3.26 mi every packet arrived; by 3.75 mi two position packets and a message in a row had failed. In between, packets were starting to lose the coin flip. Podium line: *"3.75 miles is where THIS setup stopped being reliable — the physics has several more miles in it, and the first antenna upgrade collects them."*

Q: What exactly is "modulation," and what is the encryption key?

Modulation is how the radio wave carries bits — the grammar. LoRa's scheme is chirp spread spectrum: every symbol is a tone sweeping upward across the full 250 kHz channel, and the information is in WHERE the sweep starts — SF11 gives $2^{11} = 2048$ possible start positions, so one chirp carries 11 bits. The receiver knows the sweep shape and "de-chirps," collapsing each sweep to a point; noise doesn't collapse, which is why LoRa decodes 17.5 dB below the noise floor. CR 4:5 adds one repair bit per four data bits. A receiver expecting SF11 hears an SF9 transmission as gibberish — right frequency, wrong grammar — hence modulation is row two of the contract. The encryption key is the secret — a 256-bit (32-byte) random AES key generated when the Group9 channel was created, carried inside the channel URL both radios were provisioned with over USB, never transmitted over the air (hence PSK, Pre-Shared Key). Same grammar lets you hear; same secret lets you understand. (The actual key lives in the team repo's provisioning URL — describe it in public, never display it.)